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## What is AWS VPN?

AWS VPN (Virtual Private Network) allows you to **securely connect your on-premises network or client device** to your Amazon VPC over an **encrypted connection** via the public internet or AWS Direct Connect.

Think of it as a **secure bridge between your data center (or laptop) and your AWS VPC**.

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## Types of AWS VPN

| Type             | Description                                                                 |
|------------------|-----------------------------------------------------------------------------|
| Site-to-Site VPN | Connects your <b>entire on-prem network</b> to an AWS VPC                   |
| Client VPN       | Connects <b>individual users/devices</b> (like laptops) to your AWS network |

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### ◆ 1. AWS Site-to-Site VPN

- **Use case:** Enterprise/data center to AWS VPC
- **Protocol:** IPsec (Internet Protocol Security)
- **Requires:**
  - A **Virtual Private Gateway (VGW)** or **Transit Gateway** on AWS side
  - A **Customer Gateway (CGW)** on your side (router/firewall supporting IPsec)

 **Components:**

- **VPC** — Your AWS network
- **Virtual Private Gateway (VGW)** — AWS side endpoint
- **Customer Gateway (CGW)** — On-premises side (router or firewall)
- **VPN Connection** — The encrypted IPsec tunnel

#### **Security:**

- Two **IPsec tunnels** are created for high availability
  - All traffic is **encrypted in transit**
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## ♦ **2. AWS Client VPN**

- **Use case:** Developers, remote employees, individual users
- **Protocol:** OpenVPN-based
- Connects a user's laptop directly to AWS resources (like private EC2s)

#### **Features:**

- Fully managed by AWS
- **Authentication options:**
  - Active Directory
  - Mutual certificate
  - Federated (SAML)

- You configure **routes and authorization rules** for access
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## How Does AWS VPN Work?

### Site-to-Site Example Workflow:

Your Data Center

↓ Encrypted Tunnel (IPsec)

Customer Gateway (CGW)

↔ VPN Connection

Virtual Private Gateway (VGW)

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AWS VPC (with EC2, RDS, etc.)

### Client VPN Example:

Employee Laptop

↓ Encrypted VPN (OpenVPN)

AWS Client VPN Endpoint

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Private AWS Resources (EC2, RDS, etc.)

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## Use Cases

- Securely extend your **on-premises network** to the cloud
- Allow **remote workforce** to access private AWS resources
- **Hybrid cloud** connectivity
- **Migrate workloads** from data center to AWS

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## Important Notes

| Feature                    | Site-to-Site VPN      | Client VPN               |
|----------------------------|-----------------------|--------------------------|
| Connection Type            | Network-to-network    | User-to-network          |
| Protocol                   | IPsec (IKEv1/v2)      | OpenVPN                  |
| Max Bandwidth (per tunnel) | ~1.25 Gbps (variable) | Depends on client config |
| Managed by AWS?            | Partially             | Fully managed            |
| Common Use Case            | Hybrid cloud          | Remote developer access  |

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## Alternatives to VPN

- **AWS Direct Connect** (dedicated private link)
  - **Transit Gateway** (for scaling to many VPCs/sites)
  - **AWS PrivateLink** (for private service access without VPN)
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